



Emergency Lane

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Difficulty Level : Medium

Emergency Lane

A power surge hits the GCPC control network just before the final rehearsal. Nadia must drive from the backup garage in city 1 to the main broadcast hub in city n as fast as possible.

The city road map has n intersections and m bidirectional roads. Each road has a positive travel time. To help emergency vehicles, the city grants Nadia one **priority pass**: she may use it on **at most one road**, making the travel time of that single road equal to 0.

She may choose not to use the pass at all.

Task: Find the minimum possible time to travel from city 1 to city n . If city n is unreachable, print -1.

Input

The first line contains two integers n and m :

- n — the number of cities
- m — the number of roads

Each of the next m lines contains three integers u , v , and w , describing a bidirectional road between cities u and v with travel time w .

Output

Print a single integer — the minimum possible travel time from city 1 to city n using the priority pass on at most one road, or -1 if no route exists.

Sample Input 1

```
5 6
1 2 4
2 5 9
1 3 2
3 4 2
4 5 2
2 4 3
```

Sample Output 1

4

Sample Input 2

4 2
1 2 5
3 4 1

Sample Output 2

-1

Explanation

In the first sample, without using the pass, the best route is 1 → 3 → 4 → 5 with total cost 6.

If Nadia uses the priority pass on the road (4, 5), the same route costs

$$2 + 2 + 0 = 4,$$

which is optimal.

In the second sample, city 4 is disconnected from city 1, so the answer is -1.